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The Addition Paradox

An energy thesis for Canada.

Why the energy transition did not happen, why that is acceptable, and why Canada is uniquely positioned to win the forty years that follow.

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EXECUTIVE SUMMARY

A record year for clean energy was also a record year for fossil emissions. Both are true. Both will remain true.

The most consequential energy fact of 2024 was not the record deployment of renewables. It was that the record deployment of renewables coincided with a record year for fossil-fuel combustion. Both happened. Neither cancelled the other. The global system did not transition. It *expanded*.

This paper documents that expansion, names what it implies, and argues that Canada — uniquely among industrial democracies — holds the cards to win the next forty years of a global energy market that will not, in any honest reading, be exclusively electric or exclusively clean. The recommendation is that Canadian energy policy should stop apologising for the production of energy and start designing for the export of it — not as a transitional embarrassment, but as a forty-year national project to which clean and conventional energy contribute in proportion to the demand they actually serve.

Key conclusions

1. **The global "energy transition" as conventionally described is not happening.** Investment in clean energy reached two trillion dollars in 2024, twice the level invested in fossil fuels. The share of fossil fuels in global electricity supply fell by less than two percentage points. The clean buildout has not displaced. It has added.
2. **The substitution that did occur was not from molecules to electrons, but from one chokepoint to another.** Eighty to ninety-five percent of refining and assembly capacity for the technologies of the clean transition is concentrated inside the borders of a single nation, which is also engaged in the largest naval expansion of the twenty-first century. The world swapped Saudi influence for Chinese influence and called it independence.
3. **Canada holds, simultaneously and uniquely, the four assets the next forty years will reward.** The third-largest oil reserves, with a barrel among the cleanest of any major producer; the largest hydroelectric production per capita in the G7; the second-largest reserves of high-grade uranium; and Pacific deepwater export capacity within about ten sailing days of the largest energy import markets on earth. No other G7 nation holds this combination.

Bottom line: the countries that will win the next forty years are not the countries that virtue-signalled fastest. They are the countries that supplied the world's actual demand while building, in parallel, the foundations for what comes after. Canada is one of perhaps three countries on earth in a position to do both.

It would be ungenerous to call the energy transition a fraud. It was sincere. Its publicists believed every word they published. They simply happened to be employed by a civilisation incapable of subtraction, and so the project's life work — designed to subtract — was metabolised, in the manner of a body processing an indigestible meal, into addition.

The numbers tell the story plainly. In 2024, the world deployed 858 terawatt-hours of new clean electricity, the largest single-year clean-energy build in human history. In the same year, the world burned more coal than it had ever burned, more natural gas than it had ever burned, and emitted from its power plants 14.6 billion tonnes of carbon dioxide — also more than it had ever emitted.

How does record clean deployment produce record fossil emissions? Through a fact obvious to any honest reading of the data: global electricity demand grew by 1,172 terawatt-hours that year, driven by air conditioning required to survive heatwaves the system itself had made hotter, by data centres belonging to an industry that has not yet decided what it is for, and by electric vehicles whose batteries are charged from the same grids that power the refineries producing the fuel they were intended to replace. Of the 1,172 terawatt-hours of new demand, the clean buildout provided 858. The remaining 314 were provided by precisely those parties the buildout had been retained to retire.

> *This is not failure. This is addition. The renewable-energy industry has performed exactly the task it was hired to do, at unprecedented scale and unprecedented cost-effectiveness. What it has not done, because it cannot do alone, is shrink the demand for fossil energy.*
>
> — §1, THE DIAGNOSIS

The implication for the next forty years is structural, not cyclical. Coal generated 10,613 terawatt-hours of global electricity in 2024 and is forecast to remain the largest single source of generation in the world through the mid-2030s. Natural gas signed export contracts in 2024 with delivery windows extending past every climate target nominally in force. Oil received 570 billion dollars in upstream investment last year, the highest level since 2017, indicating that whatever rumours had reached the family of an impending inheritance, the patriarchs are not yet packing.

The deceased — the energy transition as advertised — was buried with a thick stack of renewable-energy certificates. They will be of considerable comfort in the dark.

2. THE CHOKEPOINT MIGRATION

The honest reframing of the past decade is not that the world replaced fossil energy. It is that the world replaced one form of energy concentration with another, and the new concentration is, by any sober reading of geopolitical risk, more dangerous than the one it succeeded.

The chokepoints of the molecule economy were known. Saudi crude. Russian gas. Hormuz, Suez, Panama, the Bosporus. These were the places where energy supply could be physically interdicted. They were defended by alliance structures built over seventy years, by a global naval order, and by mutual interest in their continued open functioning. The chokepoints of the mineral economy are also known. The defence is not yet built.

Polysilicon, the foundational material of solar photovoltaics, is refined for approximately ninety-five percent of global supply by facilities concentrated inside the borders of the People's Republic of China. Lithium battery cells, the foundational technology of grid storage and electric vehicles, are manufactured for approximately eighty percent of global supply by the same nation. The neodymium-iron-boron permanent magnets required for direct-drive wind turbines and electric-vehicle traction motors are sourced for upwards of ninety percent of global supply from facilities within the same political jurisdiction.

This is not an artefact of comparative advantage. It is the result of two decades of explicit national industrial policy, executed with patience, capital, and an absence of the regulatory frictions that prevent comparable buildouts in democracies. The clean-energy supply chain is, on every objective measure, more concentrated than the oil and gas supply chains of the 1970s. The same nation that controls those supply chains is engaged in the largest naval shipbuilding programme of the twenty-first century. The number of warships commissioned by the People's Liberation Army Navy in the past five years exceeds the entire active fleet of any European NATO member. The strategic significance is not contested.

> *Every megawatt of solar, wind, or battery storage deployed in a democracy increases that democracy's dependence on a nation whose strategic objectives are explicitly revisionist. The transition did not buy independence. It bought a different dependence.*
>
> — §2, THE CHOKEPOINT MIGRATION

This is not an argument against clean energy. It is an argument against pretending that clean energy as currently sourced is independent energy. Honest decarbonisation requires a supply chain that does not depend on the strategic permission of a single adversary. The transition will not be honest, or durable, until that condition is met.

3. THE CANADIAN THESIS

The temptation for Canadian policymakers reading the global energy picture is to mourn it. Canada is a major fossil producer; the world is being told it must consume less fossil energy; ergo Canada is on the wrong side of history. This reading is wrong on the facts, wrong on the timing, and wrong on the strategy. Canada is the only G7 nation that holds, simultaneously, all four of the assets the next forty years of global energy will reward.

First asset · Among the cleanest barrels of the commodity the world will continue to consume

Canada holds the world's third-largest oil reserves, produced with a barrel among the cleanest of any major producing nation. The emissions intensity of Canadian oil-sands production has declined by thirty-three percent since 2009, with the largest operators now producing crude at carbon intensities below the global average. New in-situ projects approved this decade target emissions intensities below sixty percent of the conventional benchmark. The country that the loudest voices in the climate movement have spent fifteen years asking the world to boycott is, on the evidence, among the cleanest major producers of the commodity the world will continue to consume for the next four decades.

Second asset · The largest clean electricity production per capita in the G7

Canadian hydroelectric generation exceeds 380 terawatt-hours annually, with sixty percent of installed capacity located in jurisdictions — Quebec, British Columbia, Manitoba, Newfoundland and Labrador — with surplus that can be exported, either as electrons across the U.S. border or as molecules in the form of green hydrogen, ammonia, or processed metals manufactured with clean power.

Third asset · The highest-grade uranium reserves on earth, located inside a stable democracy

Canada holds the world's second-largest reserves of high-grade uranium, concentrated in Saskatchewan's Athabasca Basin, with grades on average ten to one hundred times higher than the global mean. As the small modular reactor industry transitions from regulatory development to commercial deployment, the country that produces the fuel will hold a strategic position comparable to that of Saudi Arabia in the 1970s oil market — except that this resource is located in a stable democracy, governed by transparent law, with established regulatory institutions and a hundred-year history of safe operation.

Fourth asset · Pacific and Atlantic deepwater export capacity

Canadian deepwater Pacific ports — Kitimat, Prince Rupert, Vancouver — sit within about ten sailing days of Yokohama, Busan, and Shanghai. The same Atlantic seaboard infrastructure that built the modern North Atlantic trade can serve a re-industrialising Europe seeking to escape dependence on adversarial gas supplies. Taken with the three assets above, no other G7 nation holds this combination.

What does Canada do with this combination?

It exports. Not apologetically. Not transitionally. As a forty-year national project. The architecture of that project is the subject of the companion papers in this series — the Canada–Korea Pacific Infrastructure Facility (WP1, *The Bilateral Foundation*), the Canada–United States Energy and Compute Compact (WP2), and the Canada–Korea

Pacific Defence-Industrial Corridor (WP3). The premise of all three, and the premise defended here, is that Canada's energy future is not a problem to be managed but a generational opportunity to be seized.

To seize it requires the abandonment of one specific intellectual habit: the habit of treating fossil-fuel production as a moral failure to be apologised for. The fossil fuels Canada produces will be produced and consumed somewhere; the only question is whether they will be produced in Canada, under Canadian labour and environmental standards, with Canadian carbon-capture investment and Canadian Indigenous economic participation, or elsewhere under conditions worse on every measurable dimension. The continued displacement of Canadian production by less responsible production in other jurisdictions does not reduce global emissions. It increases them. It also impoverishes Canada.

> *The countries that will win the next forty years are not the countries that virtue-signalled fastest. They are the countries that supplied the world's actual demand while building, in parallel, the foundations for what comes after. Canada is one of perhaps three countries on earth in a position to do both.*
>
> — §3, THE CANADIAN THESIS

4. THE VERDICT

The deceased — the energy transition as advertised — is alive, well, and presently boarding a private jet to the next summit, where attendance is mandatory, the food, as always, is excellent, and the bill, as always, will be settled later.

What is not alive is the comfortable story it told. The story was that virtue and prosperity were aligned: that the fastest emitters were the worst people, that the cleanest producers would be rewarded, that the supply chains of the future would be more democratic than the supply chains of the past. None of this turned out to be true. The fastest emitters were the people whose populations were industrialising, whose grids could not yet carry the load, whose air conditioning was a survival good rather than a luxury. The cleanest producers — and Canada is among the cleanest — were boycotted by the people who continued buying from dirtier producers because

the dirtier producers were closer or cheaper or both. The supply chains of the future turned out to be more concentrated, and more authoritarian, than the supply chains they replaced.

These are the facts. They are not pessimistic facts. They are honest ones.

Canada has spent fifteen years arguing about whether it should be ashamed of its energy industry. While that argument was occurring, the global energy system was settling into a new geometry. In that geometry, Canada holds among the cleanest barrels of the commodity the world will continue to consume for forty more years, the largest per-capita supply of zero-carbon electrons in the G7, the dominant share of the highest-grade uranium reserves on earth, and Pacific and Atlantic deepwater export capacity — a combination no other G7 nation holds.

A country with this hand does not need to apologise. It needs to play. The next forty years of global energy will be won by the producers who can deliver responsibly, at scale, into the markets that will continue to need molecules and into the markets that will increasingly need clean electrons, while building the supply chains and alliances that will define the energy order of 2065. This is Canada's hand. It has rarely been a better one.

> *The casket will remain closed.*
>
> — JESSE JAMES · NORTH PACIFIC STRATEGY INITIATIVE · 15 MAY 2026

SOURCES

Added at v1.0.1, 14 October 2026: one source line per section, listing only sources retrieved and checked for this edition. Figures with no line here had no source located and are stated as gaps.

- **§1 · The Diagnosis.** Ember, *Global Electricity Review 2025*, 8 April 2025: renewables added a record 858 TWh of generation in 2024; power-sector emissions rose to an all-time high of 14.6 billion tonnes of CO_2 (ember-energy.org <https://ember-energy.org/latest-insights/global-electricity-review-2025/>). International Energy Agency, "Investment in clean energy this year is set to be twice the amount going to fossil fuels," 6 June 2024 (*World Energy Investment 2024*): some $2 trillion to clean technologies against slightly over $1 trillion to coal, gas and oil; upstream oil and gas investment of $570 billion in 2024 (copy hosted by the US Department of Energy <https://www.energy.gov/sites/default/files/2024-06/125.%20IEA%2C%20Clean%20Energy%20Investment.pdf>). Not located for this edition: the 1,172 TWh demand-growth total and its 858/314 split, and the 10,613 TWh coal figure.

- **§2 · The Chokepoint Migration.** No source was retrieved and checked for this edition. The polysilicon, battery-cell and permanent-magnet shares and the naval comparison are stated without a cited source.
- **§3 · The Canadian Thesis.** Natural Resources Canada, “Oil resources” (modified 16 January 2025): Canada has the third-largest proven oil reserve in the world, an estimated 171.0 billion barrels (natural-resources.canada.ca <https://natural-resources.canada.ca/energy-sources/fossil-fuels/oil-resources>). S&P Global Energy, 24 June 2026: oil-sands greenhouse-gas intensity fell nearly one-third (31 percent) between 2009 and 2025 (press.spglobal.com <https://press.spglobal.com/2026-06-24-Canadian-Oil-Sands-Greenhouse-Gas-Intensity-Continues-a-17-year-Decline-Down-Nearly-One-third-Since-2009,-S-P-Global-Energy-Analysis-Finds>). Canada Energy Regulator, Canada energy profile: 625.7 TWh generated in 2021, 60 percent of it hydroelectric (cer-rec.gc.ca <https://www.cer-rec.gc.ca/en/data-analysis/energy-markets/provincial-territorial-energy-profiles/provincial-territorial-energy-profiles-canada.html>). World Nuclear Association, “Uranium in Canada”: the majority of Canada’s uranium resources are in high-grade deposits, “some one hundred times the world average” (world-nuclear.org <https://world-nuclear.org/information-library/country-profiles/countries-a-f/canada-uranium>). Not located for this edition: the in-situ intensity targets, the uranium-reserve ranking, the surplus share of hydro capacity, and sailing times.
- **§4 · The Verdict.** Restates §1–§3; sources as above.

ENGAGE WITH THIS WORKING PAPER

Substantive comment, technical critique, and named response are welcomed. The Working Paper is maintained as a public version-controlled document on GitHub; the version above is the editorially-controlled v1.0.1.